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YOU MUST HAND-WRITE YOUR ANSWER

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF CHEMISTRY

CHEM1043 General Chemistry II

December 8, 2020 (Tuesday) **9:30 am – 11:30 am**

(Cutoff time for uploading your answer script to OLEX: 12:00 noon)

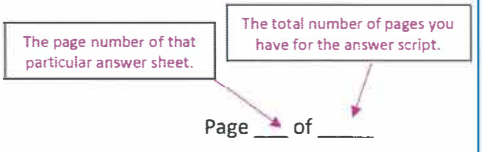
Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

This paper consists of **six** questions (100 marks). Answer **ALL** questions.
Hand-write your answers on A4 size white papers using black pen or blue pen.
Write on each answer sheet your student number and the page number.

A list of physical constants and a periodic table are provided on the last pages of this question paper.

At the end of the examination, please do the following:

- ☐ Check that you have written your **student number** and **page number** on each page of your answer script.



- ☐ Scan or take photos of your answer script and combine the pages into one single file.
- ☐ Check your file to make sure all pages are included and the images are clear.
- ☐ Submit your file through OLEX. And **make sure you have submitted the correct and complete file**.

There are **six questions** in this paper. Answer **ALL** questions. **Hand-write** your answers on A4 paper.

Question 1 (17 marks)

(a) (i) Compound *X* has a boiling point well below 100 °C. A 250.0 mL flask has a mass of 47.461 g when evacuated and a mass of 47.926 g when filled with the vapour of *X* at 100 °C and 745 mmHg. Calculate the molar mass of *X*. Show your calculations.

(4 marks)

(ii) A volume of 27.0 mL of nitrogen gas was collected in a tube at a temperature of 18.0° C and a pressure of 750 mmHg. The next day the volume of the nitrogen was found to be 27.6 mL with the barometer still reading 750 mmHg. What was the temperature on the second day? Show your calculations.

(3 marks)

(b) (i) Using the van der Waals equation, calculate the temperature of 20.0 mole of helium in a 10.0 L cylinder at 120 atm. The van der Waals equation is $P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$ where for He, $a = 0.0341$ atm L^2 mol^{-2} and $b = 0.0237$ L mol^{-1}. Show your calculations.

(3 marks)

(ii) Compare this value with the temperature calculated from the ideal gas equation. Show your calculations.

(2 marks)

(iii) In general, discuss in detail how real gases are different from ideal gases.

(5 marks)

Question 2 (18 marks)

(a) Consider Be_2^-, CO and O_2^+ .

(i) Represent the bonding with a labelled molecular orbital (MO) diagram in each case.

(ii) Predict the bond order and magnetic property (i.e. diamagnetic or paramagnetic) in each case. Explain your answer.

(iii) Compare the expected bond lengths and the relative stabilities (towards dissociation) of Be_2^-, CO and O_2^+ . Give your reasoning.

(14 marks)

Question 2 (Con't)

(b) Use band theory to explain why magnesium conducts electricity.

(4 marks)

Question 3 (19 marks)

(a) At 25 °C and an O_2 pressure of 1.00 atm, the aqueous solubility of O_2 (g) is 28.31 mL per litre of water.

(i) Calculate the molarity of O_2 in the saturated solution at 25 °C when the pressure of O_2 is 3.84 atm.

(ii) The pressure is kept at 3.84 atm, and the temperature is increased from 25 °C to 35 °C. Given that the solubility of O_2 in water is 3.12×10^{-3} M at 35 °C and 3.84 atm, what volume of O_2 (measured at 25 °C, 3.84 atm) would be expelled when we heat up 1.2 L water from 25 °C to 35 °C?

(9 marks)

(b) The vapour pressure of pure water at 100 °C is 1.00 atm. Use Raoult's law to estimate the vapour pressure of water over an aqueous solution at 100 °C containing 50.0 g of ethylene glycol, $C_2H_6O_2$ (l), dissolved in 100.0 g of water. Calculate the vapour pressure lowering for water.

(5 marks)

(c) A 4.04 g sample of compound *Z* is dissolved in enough water to make a solution of 100.0 mL. The osmotic pressure of the solution is 8.73 mmHg at 298 K. Determine the molar mass of *Z*. Show your calculations.

(5 marks)

Question 4 (16 marks)

(a) The value of K_{sp} of $Mg(OH)_2$ is 1.8×10^{-11}. Calculate the molar solubility of $Mg(OH)_2$ in

(i) pure water,

(ii) 0.0862 M $MgCl_2$ (aq),

(iii) 0.0355 M KOH (aq).

Show your calculations.

(9 marks)

(b) Suppose we mix 50.0 mL of 1.00×10^{-2} M $AgNO_3$ (aq) with 100.0 mL of 2.00×10^{-4} M K_2CrO_4 (aq) at 25 °C. Does Ag_2CrO_4 (s) precipitate from the solution? Justify your answer. If yes, calculate the amount (in mmol) of Ag_2CrO_4 (s) precipitate. Given that K_{sp} of $Ag_2CrO_4 = 1.1 \times 10^{-12}$.

(7 marks)

Question 5 (18 marks)

(a) Consider the reaction, $2\ NaHCO_3\ (s) \rightarrow Na_2CO_3\ (s) + H_2O\ (l) + CO_2\ (g)$.

Molecule	$\Delta H_f°$, kJ mol^{-1}	$S°$, J mol^{-1} K^{-1}
$NaHCO_3$ (s)	−950.8	101.7
Na_2CO_3 (s)	−1131	135.0
H_2O (l)	−285.8	69.91
CO_2 (g)	−393.5	213.7

Use the data in the table above to establish the following quantities for the reaction at 298 K:

(i) $\Delta S°$ (3 marks)

(ii) $\Delta H°$ (3 marks)

(iii) $\Delta G°$ (3 marks)

(iv) K (3 marks)

Question 5 (Con't)

(b) To establish the law of conservation of mass, Lavoisier carefully studied the decomposition of mercury(II) oxide.

$HgO\ (s) \rightarrow Hg\ (l) + \frac{1}{2} O_2\ (g)$ At 25 °C, $\Delta H^\circ = +90.83$ kJ mol^{-1} and $\Delta G^\circ = +58.54$ kJ mol^{-1}.

(i) Show that the partial pressure of O_2 (g) in equilibrium with HgO (s) and Hg (l) at 25 °C is extremely low. (3 marks)

(ii) What conditions do you suppose Lavoisier used to obtain significant quantities of oxygen?

(3 marks)

Question 6 (12 marks)

For the voltaic cell,

$Sn(s) \mid Sn^{2+}\ (0.075\ M) \parallel Pb^{2+}\ (0.600\ M) \mid Pb(s)$

Use data from Table 1 (on P.6) to determine

(i) What is E_{cell} initially? Show your calculations. (3 marks)

(ii) If the cell is allowed to operate spontaneously, will E_{cell} increase, decrease or remain constant with time? Explain your answer. (3 marks)

(iii) What will E_{cell} be when $[Pb^{2+}]$ has fallen to 0.500 M? Show your calculations.

(3 marks)

(iv) What will $[Sn^{2+}]$ be at the point at which $E_{cell} = 0.020$ V? Show your calculations.

(3 marks)

– END OF PAPER –

Table 1 Standard Reduction Potentials at 298.15 K and 1 atm

Reduction Half-Cell Reaction	Standard Reduction Potential, V
Acidic solution	
$F_2(g) + 2\,e^- \rightleftharpoons 2\,F^-(aq)$	+2.866
$O_3(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons O_2(g) + H_2O(l)$	+2.075
$S_2O_8^{2-}(aq) + 2\,e^- \rightleftharpoons 2\,SO_4^{2-}(aq)$	+2.01
$H_2O_2(aq) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,H_2O(l)$	+1.763
$Ce^{4+}(aq) + e^- \rightleftharpoons Ce^{3+}(aq)$	+1.72
$MnO_4^-(aq) + 4\,H^+(aq) + 3\,e^- \rightleftharpoons MnO_2(s) + 2\,H_2O(l)$	+1.70
$PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightleftharpoons PbSO_4(s) + 2\,H_2O(l)$	+1.69
$Cl_2(g) + 2\,e^- \rightleftharpoons 2\,Cl^-(aq)$	+1.36
$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \rightleftharpoons 2\,Cr^{3+}(aq) + 7\,H_2O(l)$	+1.32
$MnO_2(s) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons Mn^{2+}(aq) + 2\,H_2O(l)$	+1.23
$O_2(g) + 4\,H^+(aq) + 4\,e^- \rightleftharpoons 2\,H_2O(l)$	+1.229
$IO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \rightleftharpoons I_2(s) + 6\,H_2O(l)$	+1.20
$Br_2(l) + 2\,e^- \rightleftharpoons 2\,Br^-$	+1.065
$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \rightleftharpoons NO(g) + 2\,H_2O(l)$	+0.956
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.800
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.771
$O_2(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2O_2(aq)$	+0.695
$I_2(s) + 2\,e^- \rightleftharpoons 2\,I^-(aq)$	+0.535
$Cu^+(aq) + e^- \rightleftharpoons Cu(s)$	+0.520
$SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,H_2O(l) + SO_2(g)$	+0.17
$Sn^{4+}(aq) + 2\,e^- \rightleftharpoons Sn^{2+}(aq)$	+0.154
$S(s) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2S(g)$	+0.144
$2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2(g)$	0.0
$Pb^{2+}(aq) + 2\,e^- \rightleftharpoons Pb(s)$	−0.125
$Sn^{2+}(aq) + 2\,e^- \rightleftharpoons Sn(s)$	−0.137
$Fe^{2+}(aq) + 2\,e^- \rightleftharpoons Fe(s)$	−0.440
$Zn^{2+}(aq) + 2\,e^- \rightleftharpoons Zn(s)$	−0.763
$Al^{3+}(aq) + 3\,e^- \rightleftharpoons Al(s)$	−1.662
$Mg^{2+}(aq) + 2\,e^- \rightleftharpoons Mg(s)$	−2.372
$Na^+(aq) + e^- \rightleftharpoons Na(s)$	−2.714
$Ca^{2+}(aq) + 2\,e^- \rightleftharpoons Ca(s)$	−2.868
$K^+(aq) + e^- \rightleftharpoons K(s)$	−2.931
$Li^+(aq) + e^- \rightleftharpoons Li(s)$	−3.05
Basic solution	
$O_3(g) + H_2O(l) + 2\,e^- \rightleftharpoons O_2(g) + 2\,OH^-(aq)$	+1.246
$OCl^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons Cl^-(aq) + 2\,OH^-(aq)$	+0.890
$O_2(g) + 2\,H_2O(l) + 4\,e^- \rightleftharpoons 4\,OH^-(aq)$	+0.401
$2\,H_2O(l) + 2\,e^- \rightleftharpoons H_2(g) + 2\,OH^-(aq)$	−0.828

Physical Constants

Planck's constant	$h = 6.626 \times 10^{-34}$ J s molecule^{-1}
Velocity of light in vacuum	$c = 2.998 \times 10^{8}$ m s^{-1}
Avogadro's number	$N = 6.022 \times 10^{23}$ molecules mol^{-1}
Boltzmann constant	$k_B = 1.381 \times 10^{-23}$ J K^{-1} molecule^{-1}
Gas constant	$R = 8.314$ J K^{-1} mol^{-1}
	$= 0.08206$ atm dm^3 K^{-1} mol^{-1}
	$= 0.08314$ bar dm^3 K^{-1} mol^{-1}
	$= 8.314$ Pa m^3 K^{-1} mol^{-1}
Faraday constant	$F = 96487$ C mol^{-1} (of electrons transferred)
	$= 23061$ cal volt^{-1} mol^{-1} (of electrons transferred)
Elementary charge	$e = 1.602 \times 10^{-19}$ C
Mass of proton	$m_p = 1.673 \times 10^{-27}$ kg
Mass of electron	$m_e = 9.109 \times 10^{-31}$ kg
Standard gravitational acceleration	$g = 9.807$ m s^{-2}
Ionic product of water	$K_w = 1.0 \times 10^{-14}$ (at 25 °C)

Some Equations for Kinetic Theory of Gases:		
$u_{rms} = \sqrt{\frac{3RT}{M}}$	$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$	$P = \frac{1}{3}\frac{N}{V}m\overline{u^2}$

Conversion Table:	**Numerical Constants:**
Specific heat capacity of water = 1 cal K^{-1} g^{-1}	$\pi = 3.141593$
= 4.184 J K^{-1} g^{-1}	
1 erg = 10^{-7} J	$e = 2.718282$
1 G = 10^{-4} T (G = gauss, T = tesla)	$\ln x = 2.303 \log x$
1 atm = 101,325 Pa = 760 mmHg (Torr)	
1 bar = 10^5 Pa	
0 °C = 273.15 K	

The periodic table

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Hydrogen 1 **H** 1.008																		Helium 2 **He** 4.0026
Lithium 3 **Li** 6.94	Beryllium 4 **Be** 9.0122			Key: Element Name Atomic number **Symbol** Atomic weight (mean relative mass)									Boron 5 **B** 10.81	Carbon 6 **C** 12.011	Nitrogen 7 **N** 14.007	Oxygen 8 **O** 15.999	Fluorine 9 **F** 18.998	Neon 10 **Ne** 20.180
Sodium 11 **Na** 22.990	Magnesium 12 **Mg** 24.305												Aluminium 13 **Al** 26.982	Silicon 14 **Si** 28.085	Phosphorus 15 **P** 30.974	Sulfur 16 **S** 32.06	Chlorine 17 **Cl** 35.45	Argon 18 **Ar** 39.948
Potassium 19 **K** 39.098	Calcium 20 **Ca** 40.078(4)		Scandium 21 **Sc** 44.956	Titanium 22 **Ti** 47.867	Vanadium 23 **V** 50.942	Chromium 24 **Cr** 51.996	Manganese 25 **Mn** 54.936	Iron 26 **Fe** 55.845(2)	Cobalt 27 **Co** 58.933	Nickel 28 **Ni** 58.693	Copper 29 **Cu** 63.546(3)	Zinc 30 **Zn** 65.38(2)	Gallium 31 **Ga** 69.723	Germanium 32 **Ge** 72.63	Arsenic 33 **As** 74.922	Selenium 34 **Se** 78.96(3)	Bromine 35 **Br** 79.904	Krypton 36 **Kr** 83.798(2)
Rubidium 37 **Rb** 85.468	Strontium 38 **Sr** 87.62		Yttrium 39 **Y** 88.906	Zirconium 40 **Zr** 91.224(2)	Niobium 41 **Nb** 92.906(2)	Molybdenum 42 **Mo** 95.96(2)	Technetium 43 **Tc** [97.91]	Ruthenium 44 **Ru** 101.07(2)	Rhodium 45 **Rh** 102.91	Palladium 46 **Pd** 106.42	Silver 47 **Ag** 107.87	Cadmium 48 **Cd** 112.41	Indium 49 **In** 114.82	Tin 50 **Sn** 118.71	Antimony 51 **Sb** 121.76	Tellurium 52 **Te** 127.60(3)	Iodine 53 **I** 126.90	Xenon 54 **Xe** 131.29
Caesium 55 **Cs** 132.91	Barium 56 **Ba** 137.33	57-70 *	Lutetium 71 **Lu** 174.97	Hafnium 72 **Hf** 178.49(2)	Tantalum 73 **Ta** 180.95	Tungsten 74 **W** 183.84	Rhenium 75 **Re** 186.21	Osmium 76 **Os** 190.23(2)	Iridium 77 **Ir** 192.22	Platinum 78 **Pt** 195.08	Gold 79 **Au** 196.97	Mercury 80 **Hg** 200.59	Thallium 81 **Tl** 204.38	Lead 82 **Pb** 207.2	Bismuth 83 **Bi** 208.98	Polonium 84 **Po** [208.98]	Astatine 85 **At** [209.99]	Radon 86 **Rn** [222.02]
Francium 87 **Fr** [223.02]	Radium 88 **Ra** [226.03]	89-102 **	Lawrencium 103 **Lr** [262.11]	Rutherfordium 104 **Rf** [265.12]	Dubnium 105 **Db** [268.13]	Seaborgium 106 **Sg** [271.13]	Bohrium 107 **Bh** [270]	Hassium 108 **Hs** [277.15]	Meitnerium 109 **Mt** [276.15]	Darmstadtium 110 **Ds** [281.16]	Roentgenium 111 **Rg** [280.16]	Copernicium 112 **Cn** [285.17]	Ununtrium 113 **Uut** [284.18]	Flerovium 114 **Fl** [289.19]	Ununpentium 115 **Uup** [288.19]	Livermorium 116 **Lv** [293]	Ununseptium 117 **Uus** [294]	Ununoctium 118 **Uuo** [294]

*lanthanoids	Lanthanum 57 **La** 138.91	Cerium 58 **Ce** 140.12	Praseodymium 59 **Pr** 140.91	Neodymium 60 **Nd** 144.24	Promethium 61 **Pm** [144.91]	Samarium 62 **Sm** 150.36(2)	Europium 63 **Eu** 151.96	Gadolinium 64 **Gd** 157.25(3)	Terbium 65 **Tb** 158.93	Dysprosium 66 **Dy** 162.50	Holmium 67 **Ho** 164.93	Erbium 68 **Er** 167.26	Thulium 69 **Tm** 168.93	Ytterbium 70 **Yb** 173.05
**actinoids	Actinium 89 **Ac** [227.03]	Thorium 90 **Th** 232.04	Protactinium 91 **Pa** 231.04	Uranium 92 **U** 238.03	Neptunium 93 **Np** [237.05]	Plutonium 94 **Pu** [244.06]	Americium 95 **Am** [243.06]	Curium 96 **Cm** [247.07]	Berkelium 97 **Bk** [247.07]	Californium 98 **Cf** [251.08]	Einsteinium 99 **Es** [252.08]	Fermium 100 **Fm** [257.10]	Mendelevium 101 **Md** [258.10]	Nobelium 102 **No** [259.10]